



Genesis of the Linghou Cu–Zn polymetallic deposit, South China: Constraints from geological, mineralogical, geochronological and sulfur isotope data

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ABSTRACT

Stratiform Cu deposits represent significant source of copper worldwide and have long been the subject of extensive scientific research. However, their origins remain highly debated. To address this key issue, we carried out a comprehensive investigation of the Linghou deposit in South China—a typical stratiform Cu deposit—utilizing microanalytical mineralogy and in situ sulfur isotope analysis of the main ore minerals, together with high-precision geochronology of genetically related hydrothermal apatite.

Hydrothermal apatite spatially and temporally associated with copper mineralization in the Linghou stratiform ores yields a lower intercept U–Pb age of 161.0 ± 4.6 Ma (MSWD = 1.09), consistent with ages obtained from the granodiorite porphyry in the deposit. Microtextural and thermometric analyses of coexisting chalcopyrite and sphalerite indicate formation temperatures of 255–303 °C, characteristic of magmatic-hydrothermal systems. The sulfur isotope compositions of sulfide minerals exhibit a narrow $\delta^{34}\text{S}$ range of +0.68 ‰ to +2.80 ‰, further supporting a genetic relationship with granitic magmatism.

Integrated geological evidence indicates that the Linghou deposit represents a Late Jurassic Cu-polymetallic skarn system. These findings not only resolve the long-standing genetic controversy of this deposit type but also highlight the considerable exploration potential for magmatic-hydrothermal Cu-polymetallic systems along the Qin-Hang metallogenic belt. The combined application of high-precision geochronology, microscale mineral chemistry, and detailed metal source tracing provides critical constraints for understanding ore-forming mechanisms, refining genetic classifications, and establishing robust metallogenic frameworks.

1. Introduction

Stratiform orebodies represent a significant host for Cu-polymetallic deposits worldwide and have long attracted significant attention. However, considerable controversy persists regarding their genetic mechanisms (Zhu et al., 2015; Zhou et al., 2022), primarily centers on the dichotomy between exhalative sedimentary (SEDEX/VMS) and magmatic-hydrothermal origins. This debate stems fundamentally from the lack of high-precision geochronological constraints, which hinders definitive correlation between mineralization age and tectonic setting. Such limitations critically impede our understanding of ore-forming processes, metal sources, and metallogenic dynamics in stratiform copper polymetallic systems, consequently compromising both ore genesis models and exploration targeting.

Recent advances in in-situ analytical techniques—including

LA-ICP-MS, MC-ICP-MS, and SIMS—have facilitated high-resolution geochronology and microscale mineralogical studies, providing new insights into mineralization processes and ore genesis (Liu et al., 2013; Hastie et al., 2020; Chinnasamy et al., 2021). Apatite, a common accessory mineral in igneous and ore systems, has emerged as a powerful chronometer for precisely constraining mineralization ages (Ma et al., 2023; Xing et al., 2025; Zheng et al., 2026). While high-precision geochronology is crucial for deciphering the genetic mechanisms of stratiform orebodies, its accuracy depends on detailed field investigations and accurate identification of hydrothermal accessory minerals spatially and temporally associated with ore minerals.

Chalcopyrite, as the predominant ore mineral in copper deposits, has attracted growing research attention for its trace element characteristics and occurrence modes (George et al., 2018; Zhao et al., 2021). Research demonstrates that its trace element distribution and morphological

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characteristics record critical physicochemical conditions during mineralization, providing key insights into ore-forming processes (Djon and Barnes, 2012; Xie et al., 2020; Tang et al., 2025). For instance, elevated selenium concentrations correlate with high-temperature hydrothermal crystallization conditions (Maslennikov et al., 2009; Chowdhury et al., 2020).

Sphalerite, the dominant ore mineral in Pb–Zn deposits, hosts various trace elements including Fe, Mn, Co, Cd, Ga, Ge, and In (Cook et al., 2009; Ye et al., 2011; Xing et al., 2022). The distribution and abundance of these elements vary with deposit type and can provide critical genetic information (Bertrandsson Erlandsson et al., 2024; Meng et al., 2024; Zeng et al., 2025). Furthermore, *in situ* sulfur isotope analysis of sulfide minerals enables precise determination of sulfur isotopic compositions at the microscale, offering robust constraints on the sources of ore-forming materials and providing compelling evidence for elucidating deposit genesis.

The Linghou Cu polymetallic deposit, located in the northeastern segment of the Qin-Hang metallogenic belt, exhibits favorable mineralization conditions and substantial exploration potential. The deposit has been continuously mined for over 50 years, with previous studies

conducting comprehensive investigations into its geological characteristics, ore-forming fluids, material sources, genetic classification, and geochronology (Tang et al., 2015, 2017; Chen et al., 2017a, b). Nevertheless, significant controversies persist regarding its metallogenic epoch and genesis, including a Hercynian sedimentary exhalative (SEDEX) deposit (Cao et al., 1988; Liu et al., 1996; Gu et al., 2003); a Yanshanian magmatic-hydrothermal deposit (Xu et al., 1981; Zhou and Yu, 1983); a skarn deposit (Tang et al., 2015; Chen et al., 2017a), and a composite model involving submarine hydrothermal sedimentation overprinted by Early Cretaceous hydrothermal activities (Chen et al., 2013).

The integration of detailed mineralogy, high-precision geochronology, and *in situ* isotopic analysis of ore and genetically related minerals offers an effective way to constrain the sources and metallogenic mechanisms of stratiform Cu-polymetallic deposits. This study investigates the Linghou Cu–Zn polymetallic deposit through high-precision U–Pb dating of hydrothermal apatite, detailed mineralogical analysis, and *in situ* sulfur isotope studies of chalcopyrite and sphalerite. The research aims to (1) precisely determine the mineralization age, (2) trace the source of ore-forming materials, and (3) elucidate the genetic

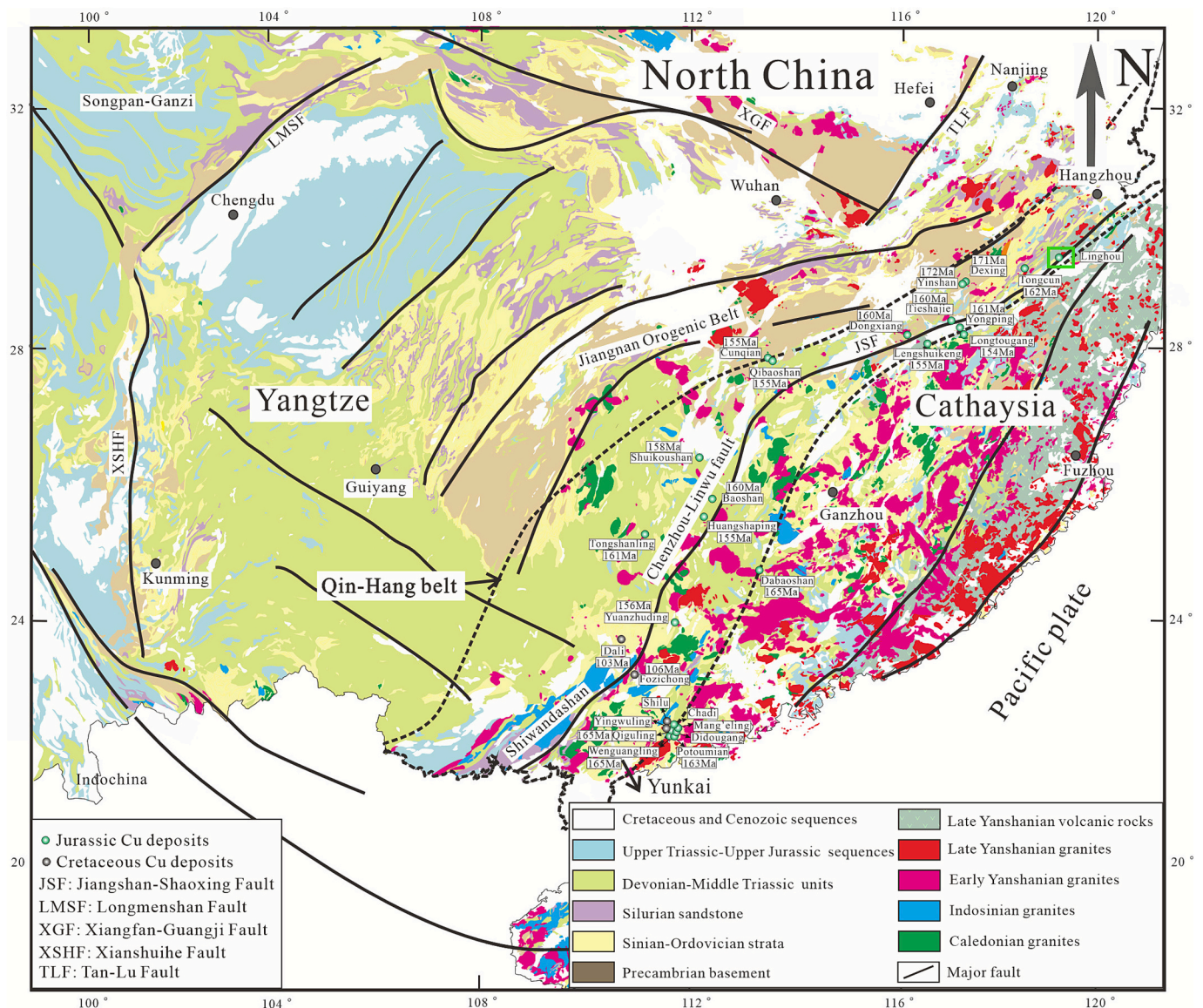


Fig. 1. Geologic map of the South China Block (modified after Zhou et al., 2006, Wang et al., 2013, and Yuan et al., 2018b). JSF: Jiangshan-Shaoxing Fault, LMSF: Longmenshan Fault, XGF: Xiangfan-Guangji Fault, XSHF: Xianshuihe Fault, TLF: Tan-Lu Fault, CLF: Chenzhou-Linwu Fault.

model of the deposit. These results will enhance the understanding of mineralization processes in comparable deposit types and provide guidance for regional mineral exploration.

2. Geological background

South China is bounded by the Qinling-Dabie-Sulu orogen to the north (Xianghuai-Guangji Fault), the Songpan-Gantze Block to the west (Longmenshan Fault), the Indochina Block to the south (Ailaoshan-Song Ma suture zone), and the Pacific plate to the east (Fig. 1; Wang et al., 2013). It was formed through the Neoproterozoic collision and amalgamation of the Yangtze and Cathaysia blocks along the Qin-Hang belt (QHB, Fig. 1), having undergone multiple tectono-magmatic events from the Proterozoic to Late Mesozoic, with superimposed influence from the Mesozoic-Cenozoic Pacific, Eurasian, and Indian dynamic systems, resulting in exceptionally complex geological structures (Ren and Chen, 1989; Zhang et al., 2013). The region is characterized by four major episodes of granitic magmatism: Neoproterozoic, Caledonian, Indosinian, and the most extensive Yanshanian granitoids (Zhou, 2003).

The QHB owes its distinctive magmatic and mineralization characteristics to its unique tectonic setting, representing South China's most significant Cu-Pb-Zn polymetallic metallogenic belt (Mao et al., 2011; Yuan et al., 2018a; Zheng et al., 2018). Gilder et al. (1996) identified a NE-trending low- T_{DM} zone composed of granitoids with elevated Sr-Nd contents and high $\epsilon_{Nd}(t)$ values through Sm-Nd and Rb-Sr isotopic analyses of Mesozoic granites in South China, designating it as the "Shi-Hang belt". Hong et al. (1999) interpreted this low- T_{DM} zone as the Proterozoic suture between the Yangtze and Cathaysia blocks, subsequently modified by multiple episodes of rifting and accretion with mantle-derived inputs. Zhou et al. (2012) subdivided the QHB into northern (eastern), central, and southern (western) segments. Although Yanshanian magmatic rocks exhibit limited surface exposure in the eastern QHB, this region displays intense magmatic activity with diverse lithologies ranging from intermediate through intermediate-felsic to felsic compositions (Lv et al., 2017). Associated mineralization is dominated by porphyry Cu-Au-Mo and hydrothermal vein-type Pb-Zn-Ag systems, exemplified by the Dexing Cu-Au-Mo deposit, the Dongxiang Cu-Au-Mo deposit, the Yinshan Cu-Au-Ag-Pb-Zn deposit,

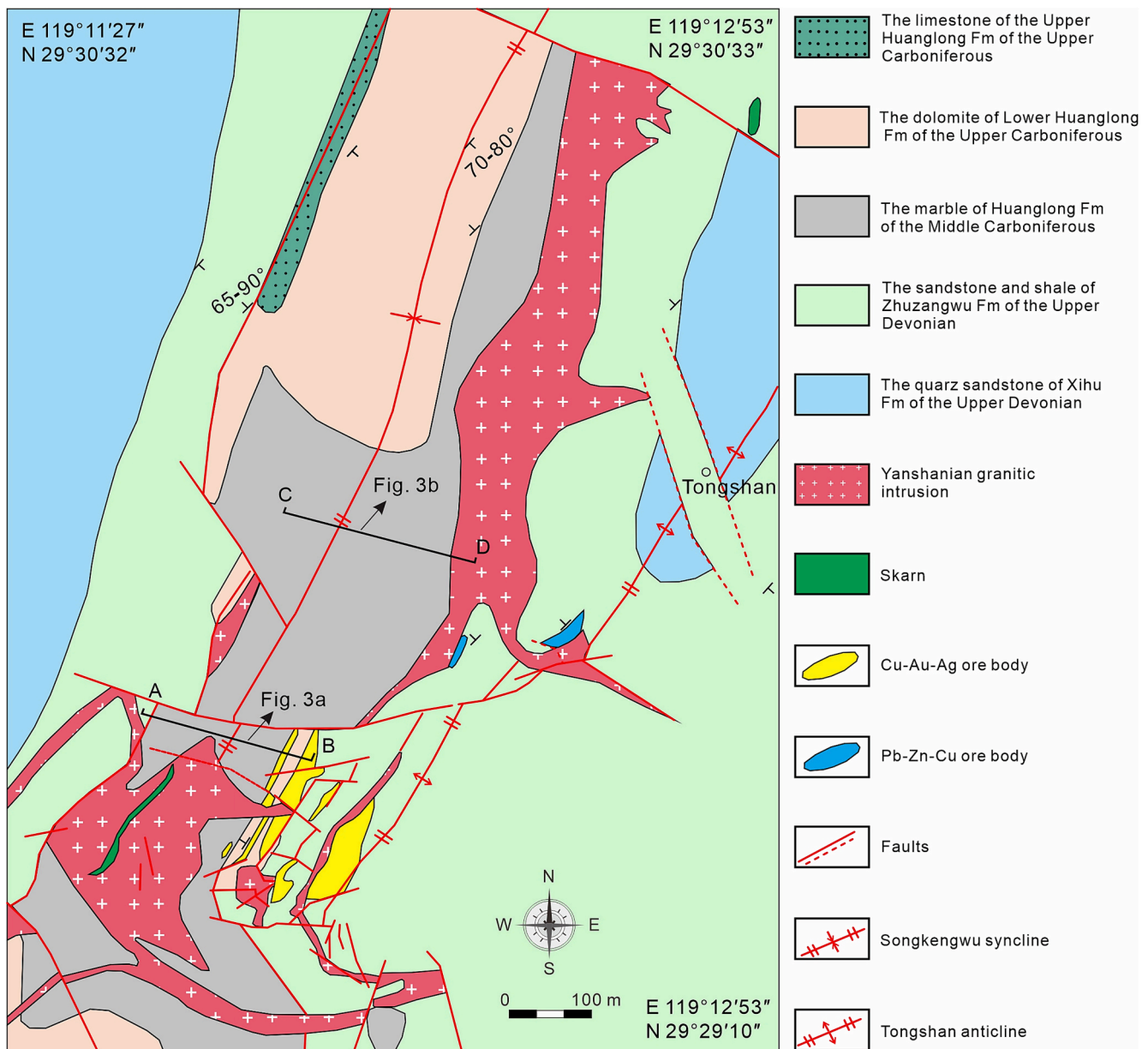


Fig. 2. Geological sketch map of the Linghou Cu-Zn polymetallic deposit (modified after Chen, 2005).

and the Linghou Cu–Zn deposit.

3. Deposit Geology

The Linghou Cu–Zn polymetallic deposit is primarily hosted in the Upper Devonian Xihu Formation (conglomeratic quartz sandstone), the Zhucangwu Formation (sandy shale and fine-grained sandstone), and the Upper Carboniferous Huanglong Formation (limestone and dolomite). The Huanglong Formation serves as the principal ore-bearing stratum (Fig. 2; Chen et al., 2017a). The mining area exhibits well-developed folds and faults. The Songkengwu syncline and Tongshan anticline form parallel composite folds with axial trends of NNE 20°–30°. The Songkengwu syncline constitutes the dominant ore-controlling structure, and fault systems are predominantly oriented NE and EW (Chen et al., 2013). The intrusive rocks are dominated by granodiorite porphyry, which occurs as dikes intruding limestone. The granodiorite porphyry exhibits a porphyritic texture, characterized by euhedral plagioclase phenocrysts set in a microcrystalline matrix, and shows a close genetic relationship with orebodies. The intrusion locally hosts disseminated chalcopyrite and pyrite, with Cu grades of 0.1–0.3%.

LA-ICP-MS zircon U–Pb dating constrains the granodiorite porphyry emplacement to 161 Ma (Chen et al., 2017b; Ni et al., 2017).

Two mineralization stages are identified in the Linghou deposit: Cu–Au–Ag and Pb–Zn–Cu (Fig. 3). The later Pb–Zn–Cu stage is commonly superimposed on the early Cu–Au–Ag stage, resulting in the formation of polymetallic orebodies (Tang et al., 2017). These orebodies are characterized by distinct veins and irregular lenses with sharp wall-rock contacts and are spatially closely related to the intrusive rocks (Tang et al., 2015; Figs. 2, 3). Mineralization in the Linghou deposit is dominated by massive ores (Fig. 4a–g). The primary ore assemblage comprises pyrite, chalcopyrite, sphalerite, galena, bornite, chalcocite, and molybdenite, with quartz, calcite, and dolomite constituting the main gangue minerals. Notably, hydrothermal accessory apatite is observed to be closely linked to massive Cu–Pb–Zn mineralization (Fig. 4g–i). The ores exhibit equigranular, brecciated, banded, and replacement textures (Chen et al., 2017c). The deposit shows intense wall-rock alteration, including skarnization, marbleization, magnetitization, silicification, and carbonatization, with marbleization and skarnization showing the closest spatial association with copper mineralization. On the basis of mineralogical assemblages and cross-

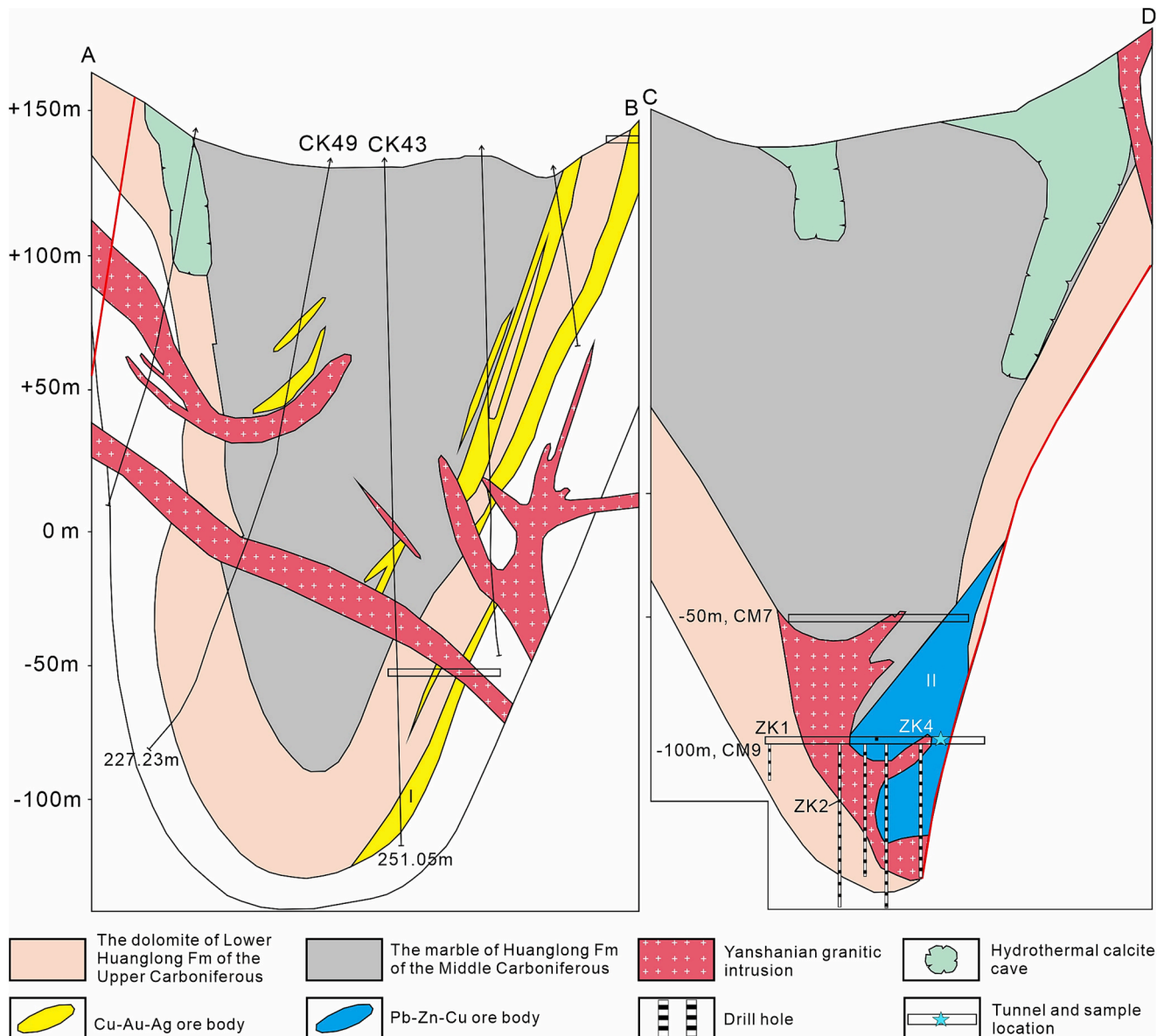


Fig. 3. Geological cross-section of exploration lines A–B and C–D from the Linghou deposit (modified after Chen, 2005).

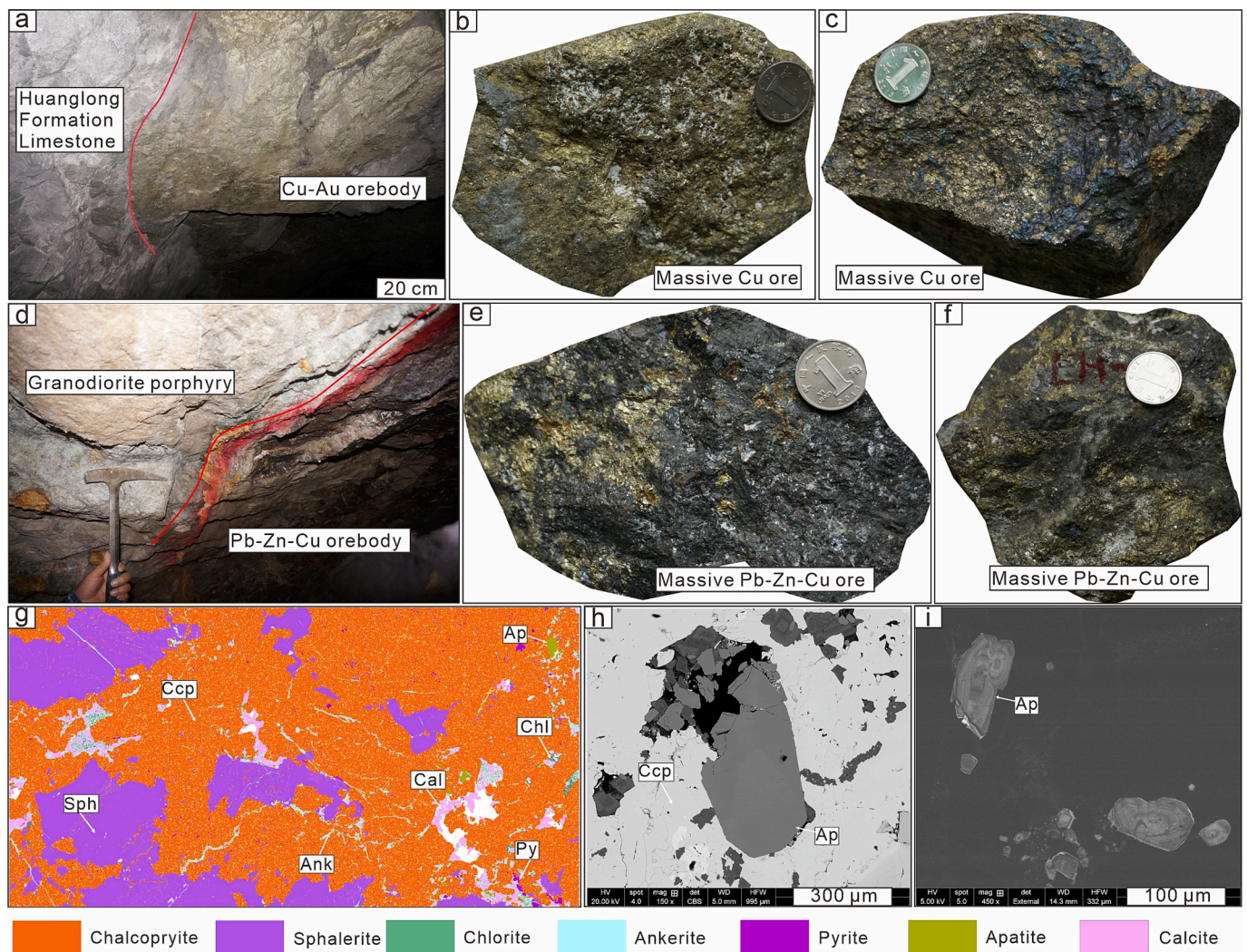


Fig. 4. Representative Cu mineralization and ore types of the Linghou deposit. (a-c) Massive Cu-Au/Cu ore; (d-f) Massive Pb-Zn-Cu ore; (g-i) TIMA map of massive copper ore, showing the close association of hydrothermal apatite with chalcopyrite, and sphalerite. Abbreviations: Py = pyrite, Ccp = chalcopyrite, Chl = Chlorite, Sph = Sphalerite, Ank = Ankerite, Ap = apatite, Cal = Calcite.

cutting relationships, the mineralization process can be subdivided into four stages: (I) Early calc-silicate (skarn) alteration stage (wollastonite + garnet + diopside + serpentine + chlorite ± epidote); (II) Cu-Au-Ag mineralization stage (quartz ± sericite + pyrite + chalcopyrite + bornite + Au-Ag minerals ± magnetite ± molybdenite); (III) Pb-Zn-Cu mineralization stage (calcite + dolomite + sphalerite + pyrite + chalcopyrite + galena); and (IV) Post mineralization stage (calcite ± quartz veins) (Tang et al., 2015).

4. Sampling and analytical Methods

Representative samples of massive Cu-Zn ore were collected for comprehensive geochronological, geochemical, and isotopic studies. Apatite cathodoluminescence (CL) and backscattered electron (BSE) images were generated using an FEI Nova NanoSEM-450 scanning electron microprobe with energy-dispersive X-ray (EDX) and Gatan MonoCL-4 detectors at the Institute of Geology, Chinese Academy of Geological Sciences (CAGS), Beijing.

4.1. Electron probe microanalysis (EPMA)

EPMA data were acquired using a JEOL JXA-8230 Electron Probe Microanalyzer at the Key Laboratory of Metallogeny and Mineral

Assessment, Institute of Mineral Resources, CAGS, Beijing. The operating conditions were 15 kV accelerating voltage and a beam current of 20 nA, with counting times between 10 s and 20 s and between 5 s and 10 s in the peak and background, respectively. The beam diameter was 5 μm, and a ZAF matrix correction was applied during the data reduction process.

4.2. Laser ablation inductively coupled-plasma mass spectrometry (LA-ICP-MS)

Chemical composition analyses of chalcopyrite and sphalerite were conducted using LA-ICP-MS at Nanjing FocuMS Technology Co. Ltd. The system consists of a 193 nm ArF excimer laser ablation unit (Analyte Excite, Teledyne Cetac Technologies) coupled to a quadrupole ICP-MS (Agilent 7700x, Agilent Technologies). The deep ultraviolet laser beam was homogenized and focused onto sulfide surfaces with a fluence of 3.0 J/cm², using a 40 μm spot diameter at 5 Hz repetition rate for 40 s ablation time. The ablated aerosol was transported to the ICP-MS by helium carrier gas. Quantitative calibration was performed using the sulfide reference material MASS-1 (USGS) and basaltic glass GSE-1G as external standards, with matrix-normalized calculations conducted without internal standardization.

4.3. LA-ICP-MS U-Pb apatite dating

Laser ablation ICP-MS U-Pb isotope dating was conducted on apatite at Yanduzhongshi Geological Analysis Laboratories Ltd, using an Analytikjena PlasmaQuant MS quadrupole with a 193 nm NWR193 Ar-F excimer laser. The MAD2 apatite standard (U-Pb TIMS age of 474.25 ± 0.41 Ma; Thomson et al., 2012) was used as an external standard for isotope fractionation correction. Additionally, the McClure Mountain apatite standard (523.51 ± 1.47 Ma; Schoene and Bowring, 2006), and MRC-1 apatite standard (153.3 ± 0.2 Ma; Apen et al., 2022) were utilized as monitoring standards. Instrument drift, mass bias and fractionation of the U-Pb ratio were corrected using a standard-sample bracketing method. The concentrations of U, Th, and Pb in apatite were determined using SRM610 as an external standard and ^{44}Ca as the internal standard element, assuming a stoichiometric apatite composition. For every 10 sample points analyzed, a set of standards comprising SRM610, MAD2, McClure Mountain and MRC-1 was analyzed. Each analysis of apatite commenced with a 20 s measurement of blank gas, followed by an additional 40 s of analysis time when the laser was switched on. Apatite was ablated using a spot size of 35 μm at a laser frequency of 8 Hz and a laser energy of approximately 3.6 J/cm^2 . The raw data were processed offline using ICPMSDataCal (Liu et al., 2010) and ZSKits software. Common Pb was corrected for utilizing the ^{207}Pb -based correction method proposed by Chew et al. (2014). Isoplot 4.15 was employed to calculate the U-Pb ages and determine the lower intercept ages on a Tera-Wasserburg diagram, which were taken to represent the time of formation of the common lead-bearing minerals (Chew et al., 2014).

4.4. TIMA (TESCAN Integrated mineral Analyzer)

The TIMA analyses were conducted at Guangzhou Tuoyan Analytical Technology Co., Ltd. using a MIRA3 scanning electron microscope. Prior to an analysis, the thin sections (mounts) were carbon-coated. During an analysis, the accelerating voltage was set at 25 kV, the probe current was 20 nA, and the working distance was 15 mm. The probe current and BSE signal intensity were calibrated using an automatic platinum Faraday cup procedure, and energy-dispersive X-ray spectroscopy (EDS) signals were calibrated using a manganese standard. The TIMA liberation analysis module was used during the analyses, with simultaneous acquisition of BSE images and EDS data. The X-ray counts for each point were set at 1000. The pixel spacing was 3 μm , and the spectrum step was 9 μm .

4.5. LA-MC-ICP-MS sulfur isotope analysis

Samples of sphalerite, and chalcopyrite were analyzed *in situ* their sulfur isotope compositions using a laser ablation coupled multi-collector inductively coupled plasma mass spectrometer (LA-MC-ICP-MS) with Neptune Plus MC-ICP-MS and Resolution S-155 Laser Ablation Systems at the MNR Key Laboratory of Metallogeny and Mineral Assessment, Institute of Mineral Resources, CAGS, Beijing, China. The laser ablation systems use helium as the carrier gas. The analyses were performed in single-point mode, utilizing a large laser spot (40–50 μm) and a low laser frequency (4 Hz). The laser energy density was fixed at 5.0 J/cm^2 . Signals for ^{32}S , $^{34}\text{S}/^{32}\text{S}$, $^{33}\text{S}/^{32}\text{S}$ were collected to calculate the $\delta^{34}\text{S}$ values. To ensure precision and to account for instrument drift, the standard was analyzed after every four analyses of the sample. The sulfur isotope mass fractionation was calibrated using the standard-sample bracketing method (SSB). Sphalerite samples were calibrated using sphalerite standard NBS123. Chalcopyrite samples were calibrated using chalcopyrite standard HTS4-6 (Li et al., 2020). The precision of the $\delta^{34}\text{S}$ analysis was within 0.5 ‰.

5. Results

5.1. Apatite U-Pb geochronology

Apatite U-Pb isotopic data are listed in Table 1. Apatite crystals spatially related to stratiform Cu mineralization display a wide range in U (3.4–105 ppm) and Th (0.6–56.3 ppm) abundances. Apatite from sample LH-16 yields concordant ages of 161.0 ± 4.6 Ma (lower intercept, MSWD = 1.09; Fig. 5a) and 160.6 ± 1.9 Ma (^{207}Pb -corrected weighted mean, MSWD = 0.89; Fig. 5b), based on 37 analytical spots.

5.2. Chalcopyrite composition

A total of 14 spot analyses were obtained, and the trace element data were summarized in Table 2; Fig. 6a). Chalcopyrite from the Linghou deposit displays three compositional characteristics: (1) Detectable Ti, V, Mn, Co, Ni, As, Mo, Sb, Te, Au, and Tl show limited variation with concentrations consistently < 10 ppm; (2) The concentrations of Ga, Ge, Se, Ag, Cd, Bi, and Pb exhibit wider variation but do not exceed 50 ppm; (3) The concentrations of Sn, Zn, and Cr exhibit significant variation, with maximum values reaching several hundred to several thousand ppm.

5.3. Sphalerite composition

The EMPA data of sphalerite are presented in Table 3. Electron microprobe analyses of sphalerite reveal Zn contents of 60.99–62.58 wt % (avg. 62.04%) and S contents of 33.00–33.56 wt % (avg. 33.21%), exceeding the theoretical S content (32.90%) but below the ideal Zn content (67.10%). Trace elements include Fe content (2.47–3.94 wt%), Cu (<2.92 wt%), Pb (<0.05 wt%), and Se (<0.06 wt%).

A total of 15 spot analyses were obtained, and the trace element data were given in Table 4 and Fig. 6b. Overall, the trace element composition of sphalerite from the Linghou deposit exhibits the following characteristics: (1) Relatively low but stable concentrations of Fe, Mn, and Cd. The Fe content ranges from $27,354 \times 10^{-6}$ to $35,594 \times 10^{-6}$, with a mean value of $32,089 \times 10^{-6}$; the Mn content varies between 340×10^{-6} and 737×10^{-6} , averaging 582×10^{-6} ; the Cd content ranges from $1,322 \times 10^{-6}$ to $1,595 \times 10^{-6}$, with a mean of $1,498 \times 10^{-6}$; (2) Enrichment in Ga and In. Ga and In content range from 24.41×10^{-6} to 203×10^{-6} (mean 105×10^{-6}) and 53×10^{-6} to 594×10^{-6} (mean 189×10^{-6}); (3) Depletion in Co and Ge. The Co concentrations are exceptionally low (0.07 – 0.38×10^{-6} ; mean: 0.19×10^{-6}), and Ge concentrations fall below detection limits in some spots, reaching a maximum of 3.82×10^{-6} ; (4) Highly variable concentrations of Cu, Pb, Ag, and Sn. The Cu content ranges from 445×10^{-6} to $24,959 \times 10^{-6}$ (mean $4,018 \times 10^{-6}$); the Pb content varies between 0.31×10^{-6} and $18,223 \times 10^{-6}$ (mean $1,498 \times 10^{-6}$); the Ag content ranges from 3.55×10^{-6} to 973×10^{-6} (mean 166×10^{-6}); the Sn content spans 0.55×10^{-6} to 250×10^{-6} , with a mean value of 52.62×10^{-6} .

5.4. In situ sulfur isotopes

In situ sulfur isotope analyses of sulfides from the Linghou massive Cu–Zn orebodies are presented in Table 5. Chalcopyrite exhibits $\delta^{34}\text{S}_{\text{CDT}}$ values ranging from 0.1 ‰ to 1.85 ‰ (mean = 1.01 ‰), while sphalerite yields a slightly narrow range of 0.57 ‰ to 1.61 ‰ (mean = 0.96 ‰).

6. Discussion

6.1. Timing of the magmatic and mineralization events in the Linghou deposit

The Linghou deposit contains the Cu–Au–Ag and Pb–Zn–Cu mineralization (Fig. 2). In the Cu–Au–Ag orebodies, Tang et al. (2017) identified molybdenite intergrown with chalcopyrite, pyrite, and quartz,

Table 1
LA-ICP-MS U–Pb ages of apatite closely associated with Cu mineralization in the Linghou deposit.

Sample No	207Pb/ 206Pb	1 σ	207Pb/ 235U	1 σ	206Pb/ 238U	1 σ	207Pb/ 206Pb	1 σ	207Pb/ 235U	1 σ	206Pb/ 238U	1 σ	207Pb- corrected age	1 σ	238U	232Th	206Pb	207Pb	208Pb
	Ratio		Ratio		Ratio		Age (Ma)		Age (Ma)		Age (Ma)				ppm	ppm	ppm	ppm	ppm
LH-16-01	0.56967	0.01257	5.60426	0.1579	0.07135	0.00125	4432.5	32.2	1916.8	24.3	444.3	7.5	165.7	7.8	10.8	1.58	0.782	0.443	1.06
LH-16-02	0.5917	0.00778	6.01108	0.10844	0.07368	0.00091	4487.7	19.1	1977.5	15.7	458.3	5.5	158.5	5.4	11	4.9	0.814	0.478	1.16
LH-16-03	0.44762	0.0066	3.01615	0.06031	0.04887	0.00066	4077.3	21.9	1411.7	15.2	307.6	4.1	159.8	3.5	16.4	16.9	0.775	0.342	0.843
LH-16-04	0.46176	0.00804	3.32663	0.08329	0.05225	0.00094	4123.5	25.8	1487.3	19.5	328.3	5.8	165.1	4.6	26.5	18.2	1.36	0.636	1.57
LH-16-05	0.47515	0.00995	3.29861	0.10173	0.05035	0.00114	4165.9	31	1480.7	24	316.7	7	154	5.4	80.1	56.3	3.9	1.84	4.5
LH-16-06	0.47601	0.00887	3.62684	0.08322	0.05526	0.00074	4168.6	27.6	1555.4	18.3	346.7	4.5	168.5	4.6	13.4	3.08	0.722	0.342	0.854
LH-16-07	0.48899	0.0064	3.68865	0.07089	0.05471	0.00077	4208.4	19.3	1568.9	15.4	343.4	4.7	161.3	3.8	13.1	2.91	0.698	0.338	0.794
LH-16-08	0.50047	0.01645	3.86634	0.14085	0.05603	0.00088	4242.6	48.4	1606.7	29.4	351.4	5.4	160.3	7.7	11.3	2.57	0.62	0.31	0.725
LH-16-09	0.5018	0.00782	3.89322	0.07939	0.05627	0.00074	4246.5	23	1612.3	16.5	352.9	4.5	160.4	4.3	12.9	2.86	0.706	0.352	0.834
LH-16-10	0.50678	0.01106	4.08139	0.1154	0.05841	0.00105	4261.1	32.1	1650.6	23.1	365.9	6.4	164.2	6	11.1	2.31	0.627	0.32	0.746
LH-16-11	0.5172	0.00911	4.08259	0.09224	0.05725	0.00081	4291	25.9	1650.8	18.4	358.9	4.9	156.3	4.9	13.8	27.1	0.759	0.395	1.18
LH-16-12	0.52262	0.01462	4.22337	0.14599	0.05861	0.00119	4306.3	41.1	1678.6	28.4	367.2	7.2	157.6	7.5	16.3	20	0.96	0.494	1.27
LH-16-13	0.52867	0.00943	4.42461	0.1017	0.0607	0.00088	4323.2	26.2	1717	19	379.9	5.3	160.3	5.3	10.8	10.4	0.638	0.334	0.874
LH-16-14	0.52906	0.0072	4.31407	0.08699	0.05914	0.00088	4324.3	20	1696.1	16.6	370.4	5.4	156	4.4	16.3	7.22	0.948	0.494	1.22
LH-16-15	0.56342	0.00997	5.26933	0.12186	0.06783	0.00101	4416.4	25.8	1863.9	19.7	423.1	6.1	160.8	6.1	12.6	8.96	0.84	0.465	1.17
LH-16-16	0.5777	0.00955	5.72787	0.12477	0.07191	0.00102	4452.9	24.1	1935.6	18.8	447.7	6.1	162.5	6.2	9.78	41.2	0.694	0.396	1.17
LH-16-17	0.58044	0.00765	5.70301	0.09422	0.07126	0.00071	4459.8	19.2	1931.8	14.3	443.7	4.3	159.5	5.1	12	6.15	0.833	0.48	1.17
LH-16-18	0.58097	0.00695	5.81316	0.09959	0.07257	0.00089	4461.1	17.4	1948.4	14.8	451.6	5.3	162.1	5	12.1	12.2	0.854	0.492	1.27
LH-16-19	0.58303	0.00919	6.14728	0.11997	0.07647	0.00088	4466.3	23	1997	17	475	5.3	169.6	6.3	10.4	10.3	0.777	0.448	1.14
LH-16-20	0.5833	0.00851	5.3885	0.09958	0.067	0.00076	4466.9	21.2	1883	15.8	418.1	4.6	148.6	5.2	12.6	4.7	0.853	0.492	1.19
LH-16-21	0.58426	0.0124	5.59796	0.16161	0.06949	0.00136	4469.3	30.9	1915.8	24.9	433.1	8.2	153.6	7.7	8.69	1.55	0.587	0.344	0.791
LH-16-22	0.58406	0.01221	6.09452	0.15894	0.07568	0.00118	4468.8	30.4	1989.5	22.7	470.3	7.1	167.2	8	9.78	3.75	0.723	0.419	1.03
LH-16-23	0.58559	0.0122	5.54046	0.13347	0.06862	0.00083	4472.6	30.3	1906.9	20.7	427.8	5	150.9	7.1	14.1	41.5	0.948	0.551	1.59
LH-16-24	0.58711	0.01014	6.20487	0.14037	0.07665	0.00112	4476.4	25.1	2005.1	19.8	476.1	6.7	167.6	7	9.94	14.5	0.738	0.429	1.12
LH-16-25	0.59498	0.0096	6.42997	0.125	0.07838	0.00085	4495.8	23.5	2036.4	17.1	486.4	5.1	166.6	6.6	11.2	2.66	0.856	0.504	1.22
LH-16-26	0.60452	0.00729	6.54307	0.10224	0.0785	0.00078	4518.9	17.5	2051.7	13.8	487.2	4.7	161	5.4	12.8	16.9	0.978	0.585	1.51

(continued on next page)

Table 1 (continued)

Sample No	207Pb/206Pb	1 σ	207Pb/235U	Ratio	206Pb/238U	1 σ	207Pb/206Pb	Age (Ma)	1 σ	207Pb/235U	Age (Ma)	1 σ	206Pb/238U	Age (Ma)	1 σ	207Pb-corrected age	1 σ	238U	ppm	232Th	ppm	206Pb	ppm	207Pb	ppm	208Pb	ppm
LH-16-27	0.60544	0.00931	6.63901	0.1351	0.07953	0.00106	4521.1	22.3	2064.5	18	493.3	6.3	162.6	6.7	8.76	0.728	0.684	0.41	0.971								
LH-16-28	0.61103	0.01057	6.72727	0.14466	0.07985	0.00102	4534.4	25.1	2076.2	19	495.2	6.1	159.8	7.4	8.95	1.32	0.72	0.435	1.11								
LH-16-29	0.61284	0.00836	7.19757	0.13065	0.08518	0.00102	4538.7	19.8	2136.2	16.2	527	6.1	169.2	6.6	9.46	1.36	0.812	0.494	1.21								
LH-16-30	0.61331	0.01053	6.18833	0.13579	0.07318	0.001	4539.9	24.9	2002.8	19.2	455.3	6	145.2	6.8	11	2.38	0.802	0.493	1.2								
LH-16-31	0.6155	0.01021	7.5199	0.16375	0.08861	0.00125	4545	24.1	2175.4	19.5	547.3	7.4	174.1	8	10.1	2.37	0.873	0.537	1.31								
LH-16-32	0.61734	0.00942	6.93208	0.13631	0.08144	0.00101	4549.3	22.1	2102.8	17.4	504.7	6	159	6.9	9.94	1.34	0.816	0.499	1.28								
LH-16-33	0.62273	0.00933	7.36267	0.14356	0.08575	0.00107	4561.9	21.7	2156.4	17.4	530.4	6.4	163.8	7.2	9.45	1.01	0.816	0.502	1.3								
LH-16-34	0.63607	0.00634	7.83962	0.19042	0.08939	0.00198	4592.6	14.4	2212.8	21.9	551.9	11.7	161.5	6.5	105	38.1	9.38	6.07	14.8								
LH-16-35	0.65557	0.02138	8.19656	0.33317	0.09068	0.0022	4636.3	47.1	2253	36.8	559.6	13	150.1	15.8	6.52	1.8	0.598	0.386	0.978								
LH-16-36	0.71368	0.01177	12.50889	0.34106	0.12712	0.00276	4758.5	23.7	2643.4	25.6	771.4	15.8	153	13.1	11.9	8.2	1.47	1.04	2.56								
LH-16-37	0.78008	0.01189	26.38276	0.55974	0.24529	0.00362	4885.8	21.7	3360.8	20.8	1414.1	18.7	168.5	25.3	3.37	0.571	0.801	0.617	1.53								

indicating their coeval formation. Consequently, the Re–Os dating of molybdenite, yielding a weighted mean age of 162.2 ± 1.4 Ma (model ages ranging from 160.3 to 164.1 Ma), is interpreted to represent the timing of Cu–Au–Ag mineralization. In contrast, the stratiform Cu–Zn mineralization previously lacked robust geochronological constraints. This study addresses this gap through detailed mineralogical work, which reveals that hydrothermal apatite is not only closely associated with but also locally replaces chalcopyrite and sphalerite. This paragenetic relationship demonstrates a genetic link between the apatite and the stratiform Cu–Zn mineralization at Linghou (Fig. 4g). Apatite U–Pb dating proves effective for mineralization age determination, as its closure temperature (450–550 °C; Kirkland et al., 2018) exceeds the estimated Cu mineralization conditions (290–334 °C; Chen et al., 2017c). Thirty-seven spot analyses of mineralization-associated apatite define a well-constrained lower intercept age of 161.0 ± 4.6 Ma. This age range overlaps with the molybdenite Re–Os age of 162.2 ± 1.4 Ma, collectively constraining the mineralization event to 160–162 Ma. Furthermore, zircon U–Pb geochronology of the mineralized granodiorite porphyry (Chen et al., 2017b) yielded a concordia age of 161 ± 2 Ma, demonstrating a precise temporal overlap between magmatism and Cu mineralization. This robust age establishes a Middle Jurassic (ca. 160–162 Ma) metallogenic event at the Linghou deposit.

6.2. Ore genesis and Cu–Zn polymetallic mineralization

6.2.1. Metallogenic temperature

Sphalerite trace element signatures, including concentrations of Fe, Mn, Ga, Ge, In, and elemental ratios (e.g., Zn/Cd, Ga/Ge), have been widely utilized as geothermometers to constrain crystallization temperatures (Möller, 1987; Cook et al., 2009; Belissont et al., 2014; Cao et al., 2014; Frenzel et al., 2016; Xing et al., 2021, 2022). Specifically, the Ga/Ge ratio proposed by Möller (1987) demonstrates robust thermometric reliability. Application of this ratio to Linghou sphalerite yields temperatures of ~ 262 °C to >300 °C on a log(Ga/Ge) vs. temperature diagram (Fig. 7), indicative of a medium-temperature-dominated mineralization environment. Complementary evidence arises from Linghou sphalerite Zn/Cd ratios (419–507; median = 443; Table 4), which primarily cluster within the moderate-temperature diagnostic field ($100 < \text{Zn/Cd} < 500$), with a subordinate subset extending into the high-temperature field ($\text{Zn/Cd} > 500$) (Liu et al., 1984). Moreover, the GGIMFis geothermometer (Frenzel et al., 2016) constrains mineralization temperatures to 255–303 °C (avg. 280 °C; Table 3), consistent with both Ga/Ge thermometry and Zn/Cd ratio interpretations.

The geochemical coherence extends to chalcopyrite-sphalerite pairs: Linghou chalcopyrite exhibits a strong Zn–Cd correlation ($R^2 = 0.88$; Fig. 8), with Cd/Zn ratios (0.004–0.008) comparable to coexisting sphalerite (0.002; Fig. 6b). This geochemical coherence between coexisting minerals suggests synchronized crystallization under stable physicochemical conditions (George et al., 2018), reinforcing the prevalence of medium-temperature-dominated regimes during the Linghou sphalerite formation.

6.2.2. Sulfur sources

Sulfur isotope compositions of ore-stage sulfides provide critical constraints on fluid sources and ore-forming processes. The absence of sulfate minerals in the ores of the Linghou deposit suggests the sulfide $\delta^{34}\text{S}$ values approximate the total sulfur composition of the hydrothermal fluid (Ohmoto and Rye, 1979). According to fractionation equations for S-isotopes from Ohmoto and Rye (1979), $\Delta^{34}\text{S}$ between chalcopyrite and sphalerite and equilibrated fluids are 0.69 to 1.60 ‰ and 0.38 to 2.95 ‰, respectively, using the average fluid inclusion homogenization temperature of 310 °C obtained for the Pb–Zn–Cu mineralization stage (Chen et al., 2017c). The calculated $\delta^{34}\text{S}$ values of the hydrothermal fluids are within the range of values of magmatic sulfur (0 ± 5 ‰; Seal, 2006), suggesting that the deposit formed from a magmatic

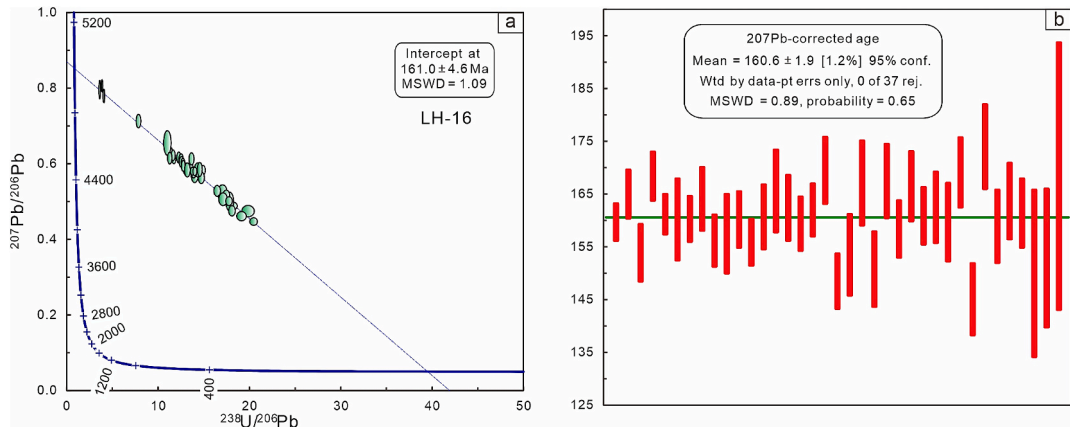


Fig. 5. Tera-Wasserburg U–Pb concordia (A) and weighted mean age (B) diagrams of hydrothermal apatite closely related to copper mineralization from the Linghou Deposit.

Table 2
Trace element concentrations (ppm) in chalcopyrite from the Linghou deposit.

Spot no.	Ti	V	Cr	Mn	Co	Ni	Zn	Ga	Ge	As	Se	Mo	Ag	Cd
LH-16-1	0.25	0.12	0.21	bdl	bdl	0.11	1092	13.27	5.53	1.66	bdl	0.43	18.06	8.26
LH-16-2	0.82	0.36	1212	0.37	bdl	2.05	3324	3.45	7.48	bdl	bdl	0.09	35.19	27.55
LH-16-3	0.36	0.14	3.96	0.44	0.09	bdl	449	13.13	11.71	1.91	3.31	0.07	20.35	3.35
LH-16-4	bdl	0.12	1.65	0.05	0.11	0.67	631	15.64	10.47	bdl	14.00	bdl	27.94	5.07
LH-16-5	1.14	bdl	1.61	bdl	0.01	0.11	858	11.66	0.73	bdl	7.01	0.17	38.09	4.72
LH-16-6	7.08	bdl	3.20	0.27	bdl	0.58	486	11.60	0.89	2.24	bdl	0.01	31.50	2.21
LH-16-7	1.13	0.11	0.09	0.65	0.11	0.72	370	7.98	58.32	bdl	5.93	0.16	40.22	1.45
LH-20-1	0.62	0.07	1.92	0.25	bdl	0.09	985	9.33	0.64	0.26	7.90	bdl	7.40	4.66
LH-20-2	0.39	0.02	1.30	0.12	0.09	0.45	1448	41.36	5.92	1.75	10.61	0.04	4.97	5.82
LH-20-3	1.57	0.08	1.78	0.32	bdl	bdl	1122	28.38	0.35	bdl	10.02	0.04	4.42	4.74
LH-20-4	0.88	bdl	bdl	bdl	bdl	0.51	1553	28.31	0.72	0.07	2.99	bdl	3.66	5.59
LH-20-5	0.54	0.07	bdl	0.17	bdl	0.13	755	15.01	0.87	bdl	4.39	0.19	4.57	3.61
LH-20-6	0.87	bdl	0.86	0.27	bdl	bdl	1053	3.02	0.67	0.35	6.18	0.11	4.89	5.34
LH-20-7	0.92	0.04	1.73	bdl	0.06	0.09	912	28.56	0.95	bdl	6.62	bdl	4.21	3.21
Min	0.25	0.02	0.09	0.05	0.01	0.09	370	3.02	0.35	0.07	2.99	0.01	3.66	1.45
Max	7.08	0.36	1212	0.65	0.11	2.05	3324	41.36	58.32	2.24	14.00	0.43	40.22	27.55
Mean	1.27	0.11	102	0.29	0.08	0.50	1074	16.48	7.52	1.18	7.18	0.13	17.53	6.11
Median	0.87	0.10	1.69	0.27	0.09	0.45	949	13.20	0.92	1.66	6.62	0.10	12.73	4.73

Spot no.	In	Sn	Sb	Te	Au	Tl	Bi	Pb	Cd/Zn	Ga/Ge	log(Ga/Ge)
LH-16-1	29.92	49.75	1.83	1.61	0.18	0.01	10.97	13.23	0.008	2.40	0.38
LH-16-2	27.09	32.26	3.06	0.05	0.96	0.02	13.60	20.45	0.008	0.46	−0.34
LH-16-3	24.85	54.21	1.62	bdl	0.05	0.03	6.17	6.47	0.007	1.12	0.05
LH-16-4	24.97	52.48	0.90	1.46	0.97	bdl	10.80	13.48	0.008	1.49	0.17
LH-16-5	28.51	73.83	0.19	2.10	1.67	0.06	5.88	6.21	0.006	16.00	1.20
LH-16-6	29.63	58.22	0.47	2.49	0.59	bdl	6.86	7.33	0.005	12.99	1.11
LH-16-7	35.02	50.62	0.82	1.00	1.17	0.01	15.35	13.06	0.004	0.14	−0.86
LH-20-1	27.83	142	1.15	4.13	0.22	bdl	10.77	2.92	0.005	14.66	1.17
LH-20-2	27.70	127	0.34	3.69	0.08	0.03	0.78	0.81	0.004	6.99	0.84
LH-20-3	27.87	129	1.91	2.26	0.08	bdl	0.83	1.03	0.004	80.36	1.91
LH-20-4	29.85	153	2.25	1.40	0.05	bdl	1.62	1.37	0.004	39.35	1.59
LH-20-5	28.11	120	1.95	2.93	0.20	0.01	4.24	3.87	0.005	17.35	1.24
LH-20-6	28.67	124	0.71	1.62	0.02	0.02	0.58	0.41	0.005	4.50	0.65
LH-20-7	27.71	148	1.18	1.50	bdl	bdl	3.20	1.41	0.004	30.05	1.48
Min	24.85	32.26	0.19	0.05	0.02	0.01	0.58	0.41	0.00	0.14	−0.86
Max	35.02	153	3.06	4.13	1.67	0.06	15.35	20.45	0.01	80.36	1.91
Mean	28.41	93.86	1.31	2.02	0.48	0.02	6.55	6.57	0.01	16.28	0.76
Median	27.99	96.70	1.17	1.62	0.20	0.02	6.03	5.04	0.00	9.99	0.98

bdl: below detection limit.

hydrothermal fluid.

6.2.3. Genesis of the Linghou Cu–Zn polymetallic deposit

Multiple genetic models have been proposed for the Linghou Cu–Zn polymetallic deposit, based on the morphology of stratiform orebodies, their spatial relationship with the granodiorite porphyry, and associated alteration-mineralization characteristics (Zhou and Yu, 1983; Xu et al., 1996; Liu et al., 1996; Liu and Cao, 1997; Gu et al., 2003; Chen et al., 2013; Tang et al., 2015, 2017).

Chalcopyrite (CuFeS₂), the dominant Cu-bearing mineral in most deposits, contains trace elements (Mn, Co, Ga, Ge, Ag, Zn, Cd, In, Sn, Sb, Mo, Se, Bi, and PGEs) that serve as discriminators for deposit classification (e.g., George et al., 2018; Wang et al., 2023; Tang et al., 2025). As shown in Fig. 9, the trace element signature of the Linghou chalcopyrite reveals that: (1) Co, Ni, Ga, Ge, As, Se, Mo, Ag, In, Sn, and Bi exhibit features strongly aligned with magmatic-hydrothermal deposits (represented by porphyry Cu deposits) but distinct from SEDEX deposits; (2) Ti, Cr, Zn, Cd, Sb, Te, and Pb concentrations show similarities to both

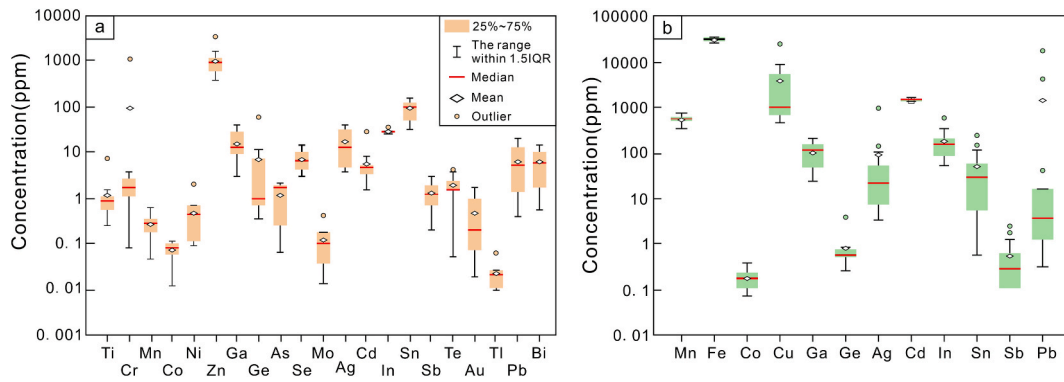


Fig. 6. Box plot of trace element composition of chalcopyrite (left) and sphalerite (right) in the Linghou deposit.

Table 3

Major element concentrations (wt%) in sphalerite from the Linghou deposit.

Spot no.	As	Se	W	Sb	Au	Ag	Pb	Sn	Fe	Co	Ni	Cu	S	Zn	Te	Cd	Total	Zn/Cd
LH-11-1	bdl	bdl	0.18	bdl	0.06	bdl	bdl	0.01	3.86	0.01	0.01	0.01	33.56	62.38	0.01	0.23	100.32	270
LH-11-2	bdl	bdl	0.12	0.05	0.10	bdl	bdl	bdl	3.88	bdl	bdl	0.02	33.20	62.32	bdl	0.24	99.92	263
LH-11-3	bdl	0.05	0.03	bdl	0.05	bdl	0.01	bdl	2.52	bdl	0.05	2.25	33.15	61.93	0.01	0.07	100.11	848
LH-11-4	bdl	0.05	0.05	0.01	bdl	bdl	0.03	bdl	3.82	bdl	bdl	bdl	33.16	62.46	0.01	0.23	99.82	268
LH-11-5	bdl	0.06	0.07	bdl	bdl	bdl	bdl	bdl	3.84	0.01	0.01	0.04	33.11	62.58	bdl	0.21	99.94	297
LH-11-6	bdl	0.04	0.12	0.01	bdl	bdl	bdl	0.01	3.76	bdl	0.04	bdl	33.28	62.48	0.03	0.21	99.97	299
LH-11-7	bdl	0.03	0.07	0.01	bdl	bdl	0.04	0.01	3.94	0.03	bdl	bdl	33.28	62.15	bdl	0.22	99.76	279
LH-11-2-1	0.01	0.02	0.10	0.03	bdl	bdl	0.05	0.03	2.72	0.00	bdl	2.92	33.15	60.99	bdl	0.10	100.12	598
LH-11-2-2	bdl	0.02	0.14	bdl	bdl	bdl	bdl	bdl	2.47	bdl	0.01	2.76	33.00	61.10	bdl	0.35	99.84	176

bdl: below detection limit.

Table 4

Trace element concentrations (ppm) in sphalerite from the Linghou deposit.

Spot no.	Zn	Mn	Fe	Co	Cu	Ga	Ge	Ag	Cd	In	Sn	Sb	Pb	Ga/Ge	Zn/Cd
LH-16-1	686,687	613	29,783	0.38	6640	65.32	0.83	52.42	1488	76	11.35	0.46	15.67	78.85	461
LH-16-2	680,800	737	35,594	0.25	702	63.89	0.54	7.75	1595	53	16.89	0.12	1.09	119	427
LH-16-3	680,734	709	35,108	0.32	840	138	3.82	21.52	1580	69	33.59	0.16	3.58	35.99	431
LH-16-4	674,708	692	35,564	0.13	862	89.01	0.69	11.69	1538	89	5.68	0.11	1.20	128	439
LH-16-5	654,727	340	33,578	0.10	24,959	42.27	0.51	141	1390	202	12.75	1.29	43.03	83.19	471
LH-16-6	679,546	655	34,750	0.23	847	45.18	0.53	10.32	1505	235	0.55	0.43	4.03	85.74	451
LH-16-7	670,182	575	31,512	0.23	5268	24.41	bdl	35.16	1322	156	1.15	0.01	9.02	—	507
LH-16-1	676,057	578	30,719	0.10	445	180	0.66	3.55	1528	98	42.55	bdl	0.31	271	443
LH-16-2	678,094	500	28,366	0.11	2634	163	0.74	21.77	1440	162	28.15	0.29	4.90	221	471
LH-16-3	672,481	438	27,354	0.12	695	116	0.27	7.66	1491	338	118	0.63	2.54	430	451
LH-16-4	667,800	523	28,878	0.18	577	47.11	0.33	3.77	1529	594	55.40	0.08	1.23	144	437
LH-16-5	660,554	603	31,943	0.14	1291	203	0.86	29.48	1534	109	58.73	0.33	3.78	237	431
LH-16-6	661,494	564	32,054	0.07	1051	168	0.57	8.43	1538	223	152	0.18	1.54	295	430
LH-16-7	653,155	637	33,797	0.24	4073	118	bdl	103	1559	220	3.74	1.83	4150	—	419
LH-16-8	637,316	559	32,334	0.21	9387	119	0.42	973	1426	210	250	2.57	18,224	285	447
Min	637,316	340	27,354	0.07	445	24.41	0.27	3.55	1322	53	0.55	0.01	0.31	35.99	419
Max	686,687	737	35,594	0.38	24,959	203	3.82	973	1595	594	250	2.57	18,224	430	507
Average	668,956	582	32,089	0.19	4018	105	0.83	95.35	1498	189	52.62	0.61	1498	186	448
Median	672,481	578	32,054	0.18	1051	116	0.57	22	1528	162	28	0.31	3.78	144	443

bdl: below detection limit; “—”: Calculation error caused by element content below the detection limit.

magmatic-hydrothermal and SEDEX deposit types; (3) Mn content differs from both deposit types, it falls closer to the characteristic range for magmatic-hydrothermal deposits. Consequently, the trace element characteristics indicate a magmatic-hydrothermal origin for the Linghou chalcopyrite.

Sphalerite is also recognized as a key indicator mineral for determining deposit type (Cook et al., 2009; Ye et al., 2011; Belissont et al., 2014; Frenzel et al., 2016; Xing et al., 2021, 2022). The analyzed sphalerite samples exhibit a considerable variation in Cu and Pb contents, with Cu concentrations ranging from 445×10^{-6} to $24,959 \times 10^{-6}$ and Pb concentrations ranging from 0.31×10^{-6} to $18,224 \times 10^{-6}$. The significant differences in the contents of these two elements among different samples reflect their highly heterogeneous distribution. During

LA-ICP-MS analysis, both Cu and Pb displayed markedly unstable signals during ablation. Combined with the commonly observed “chalcopyrite disease” texture in sphalerite and its petrographic association with galena, indicating that these elements primarily occur as mineral inclusions (e.g., chalcopyrite and galena) rather than being incorporated into the sphalerite lattice via isomorphism (e.g., Xing et al., 2021, 2022). This study employs the machine learning algorithm model of Li et al. (2023) to classify the genetic type of Linghou sphalerite, which requires complete data for all input variables and cannot process missing values. The model requires 12 input elements, including Cu and Pb. Therefore, although Cu and Pb are likely present mainly as micro-inclusions, they were fully incorporated as part of the 12 specified input elements in the genetic analysis of the deposit. Specifically, the model mandates that all

Table 5
Sulfur isotopic compositions of sulfide from massive ores in the Linghou deposit.

Sample No.	Mineral	$\delta^{34}\text{S}_{\text{V-CDT}}$ (‰)					$\delta^{34}\text{S}_{\text{H2S}}$ ^a (‰)
		Values	2 σ	Variation range	Range	Mean value	
LH-16-1	Sphalerite	0.68	0.19	0.68 ~ 1.89	1.21	1.3	1.00
LH-16-2	Sphalerite	0.98	0.18				2.95
LH-16-3	Sphalerite	1.52	0.18				2.76
LH-16-4	Sphalerite	1.41	0.18				2.13
LH-16-5	Sphalerite	1.89	0.19				0.38
LH-16-6	Chalcopyrite	0.85	0.20	0.85 ~ 2.80	1.95	2.06	0.69
LH-16-7	Chalcopyrite	2.80	0.20				1.23
LH-16-8	Chalcopyrite	2.61	0.18				1.12
LH-16-9	Chalcopyrite	1.98	0.17				1.60

V-CDT = Vienna-Canyon Diablo Troilite.

^a The mineral-H₂S fractionations were calculated using the equations in Ohmoto and Rye (1979).

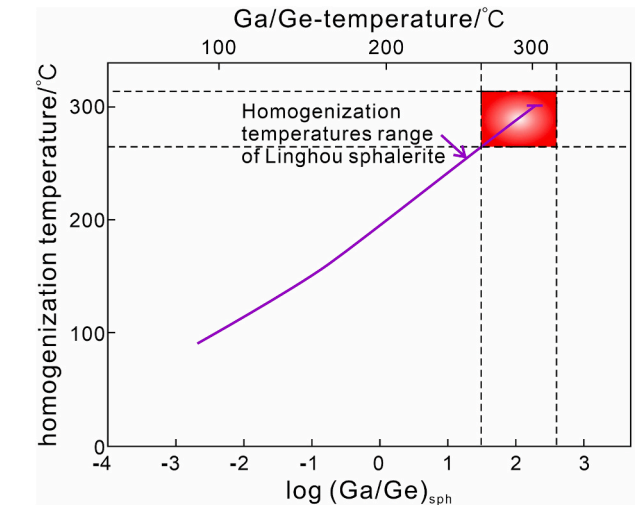


Fig. 7. The log(Ga/Ge)sph-temperature diagram showing the homogenization temperature range of the Linghou sphalerite (from Möller, 1987).

12 input elements must have concentrations above their detection limits to ensure reliable output. In accordance with this requirement, we removed three analytical points (samples LH-16-1-7, LH-16-2-1, and LH-16-2-7) from the dataset. This was because germanium (Ge) in samples LH-16-1-7 and LH-16-2-7, as well as antimony (Sb) in sample LH-16-2-1, fell below the detection limit. The remaining 12 valid data points were

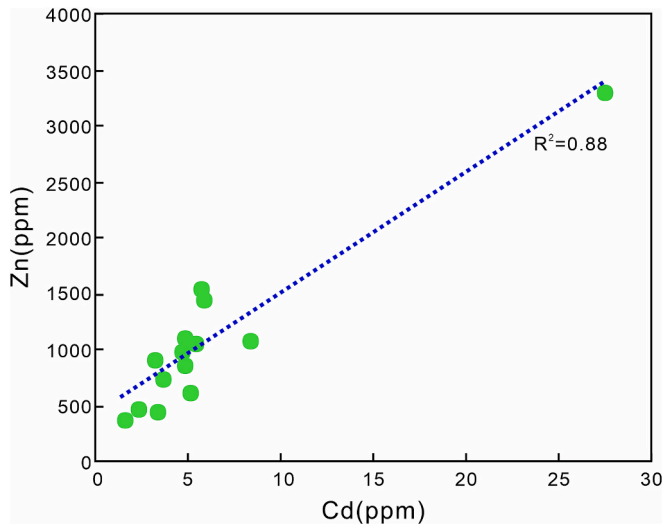


Fig. 8. Cd-Zn correlation plot of chalcopyrite from the Linghou deposit.

used for subsequent model calculations, and this selective data screening does not compromise the robustness or reliability of the overall conclusions. Results demonstrate that, excluding three test points (sample LH-16-1-7, LH-16-2-1, and LH-16-2-7) discarded due to data below detection limits, the remaining points plot within the magmatic-hydrothermal field defined by skarn and epithermal deposits, clearly distinct from the SEDEX domain (Fig. 10). All data yielded consistent genetic classification results, indicating that the presence of inclusions did not affect the conclusions drawn from the machine learning model analysis. Furthermore, we applied the model developed by Zeng et al. (2025), which integrates multiple machine learning algorithms—including random forest, extreme gradient boosting, support vector machine, and multi-layer perceptron—trained on 4908 sphalerite trace element datasets compiled from five major Zn-Pb deposit types: VMS, SEDEX, MVT, skarn, and epithermal. Results from this comprehensive model consistently classify all the Linghou sphalerite test points as magmatic-hydrothermal origin.

These findings collectively demonstrate that the Linghou deposit represents a skarn Cu polymetallic system: (1) the Cu-Au-Ag and Pb-Zn-Cu orebodies are spatially associated with the granodiorite porphyries (Figs. 2, 3, 4d), and skarn minerals and typical calc-silicate (skarn) alteration have occurred at Linghou (Xu et al., 1981; Zhou and Yu, 1983; Tang et al., 2017); (2) Our new geochronological data, integrated with previous studies, indicate that the timing of both Cu-Au-Ag and Pb-Zn-Cu mineralization (160–162 Ma) is indistinguishable from the emplacement age of the granodiorite porphyries but is significantly younger than the host rocks (Carboniferous and Devonian); (3) The Cu-Au-Ag and Pb-Zn-Cu mineralization exhibit high temperatures (300–480 °C) and moderate to high salinities (7.70–21.68 wt% NaCl eqv.) (Tang et al., 2015; Chen et al., 2017a), and S-C-H-O multi-isotopic systems consistently indicate a magmatic origin for the ore-forming fluids (Tang et al., 2015; this study). (4) Detailed microscale mineralogical studies of chalcopyrite and sphalerite further reveal a magmatic-hydrothermal origin.

6.3. Implications for regional exploration

The QHB, one of the most important Cu-polymetallic belts in eastern China, hosts numerous stratiform Cu-rich deposits. The genesis of these stratiform Cu-rich deposits has long been controversial, as exemplified by the Tieshajie, Dongxiang, Dabaoshan and Linghou polymetallic deposits investigated in this study (Fig. 1). Recent advances in micro-mineralogy and high-precision geochronology have significantly contributed to resolving these controversies. For instance, multiple evidence from the Linghou deposit studied herein support a skarn origin. The Tieshajiang Cu deposit has long been considered a Meso-Neoproterozoic VMS deposit (Wang et al., 2018). However, recent exploration revealed that polymetallic mineralization predominantly occurs along the unconformity between the Paleoproterozoic Tieshajiang Formation (greenschist facies metamorphic rocks) and Jurassic

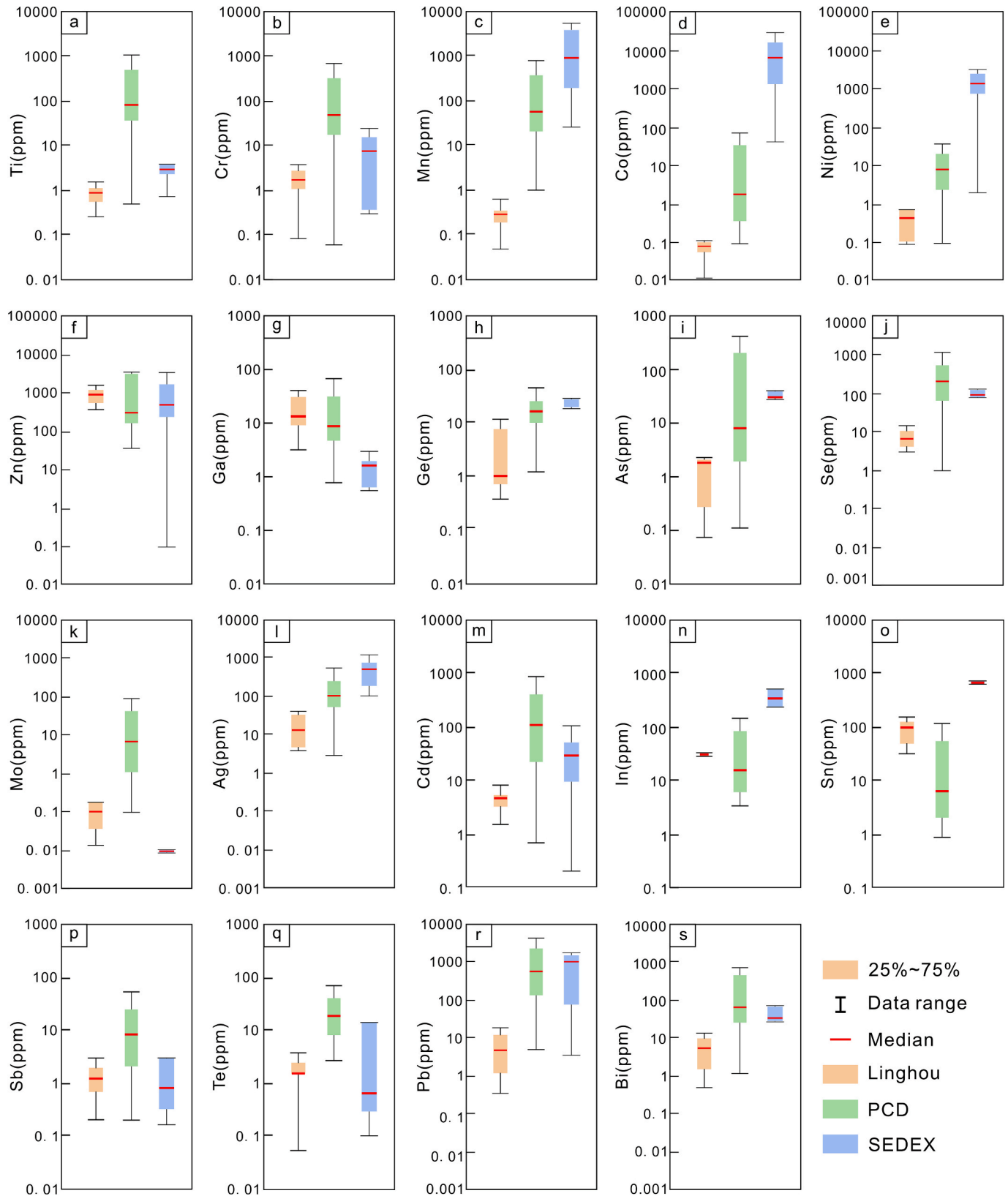


Fig. 9. Box-and-whisker plots of the trace element results of chalcopyrite from the Linghou, porphyry copper deposit (PCD), and SEDEX deposits. Data source is from Tang et al. (2025) and references therein.

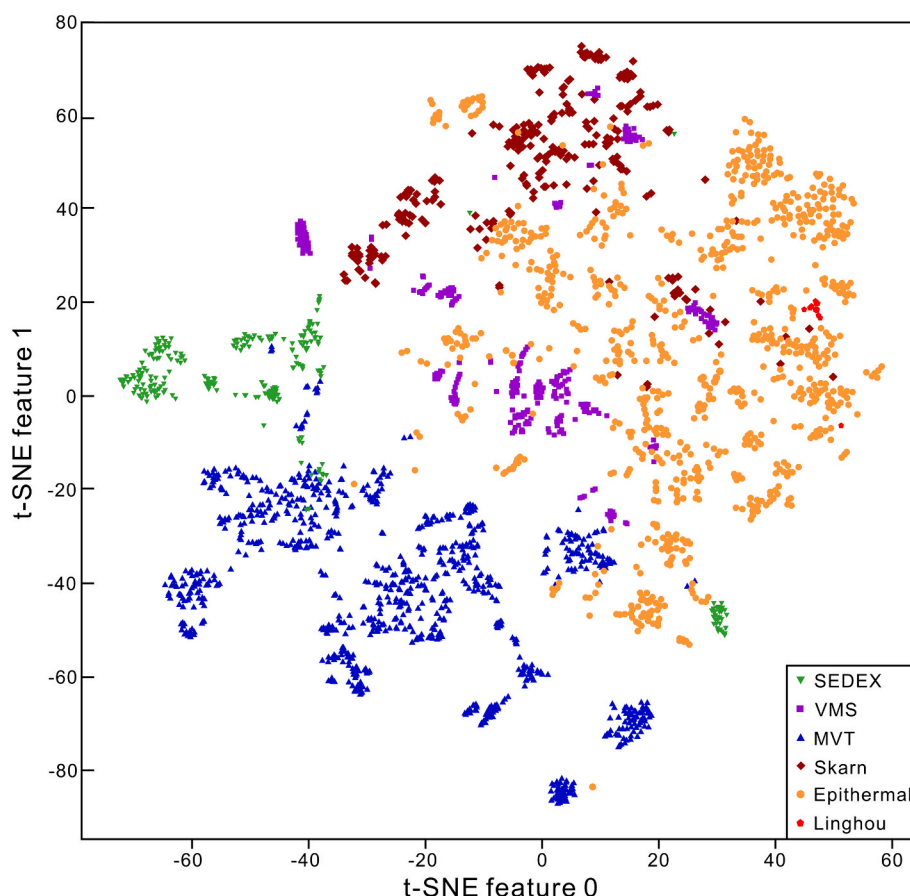


Fig. 10. T-sne plots comparing the sphalerite dataset of the linghou Cu–Zn polymetallic deposit with datasets from other deposit types (from Li et al., 2023).

Shuibei Formation. The Cu-polymetallic mineralization typically associates with quartz to form densely disseminated/massive ores or occurs as sulfide (chalcopyrite, pyrite)-quartz veinlets, exhibiting magmatic-hydrothermal features. The U–Pb dating of hydrothermal titanite coexisting with chalcopyrite yielded an age of 160.1 ± 4.4 Ma, indicating a Middle-Late Jurassic mineralization age. *In situ* S and Cu isotopes further demonstrate that the ore-forming fluids were derived from Yanshanian granites. These findings suggest the Tieshajiag deposit significantly differs from conventional VMS deposits and represents a magmatic-hydrothermal polymetallic system formed during the Middle-Late Jurassic (Jiang et al., 2024). Zheng et al. (2026) established robust temporal constraints for the Dongxiang Cu deposit through integrated geochronological approaches: Re–Os dating of molybdenite yielded a precise mineralization age of 159.6 ± 0.8 Ma, while apatite U–Pb geochronology constrained stratabound Cu mineralization to 156.4 ± 1.5 Ma. Concordant zircon U–Pb ages for associated granodiorite (157.5 ± 0.58 Ma) and granite porphyry (153.9 ± 0.63 Ma) further confirm Late Jurassic magmatic-hydrothermal activity. Combined S–Cu–Pb–H–O isotopic evidence supports the Dongxiang is a porphyry-distal hydrothermal Cu-polymetallic deposit. Furthermore, the ore-bearing intrusions in the QHB are geochemically similar to typical subduction-related continental arcs, being enriched in LILEs (Rb, Th, U, Ba, K) and LREEs but depleted in HFSEs (Nb, Ta, P, Zr, Ti) (Pearce et al., 1984; Mao et al., 2021). Consequently, recent exploration breakthroughs and associated studies in the QHB have revealed that Middle-Late Jurassic (175–155 Ma) magmatic-hydrothermal Cu–Mo–Pb–Zn–Au deposits represent the most prospective exploration targets (Zheng et al., 2026).

7. Conclusions

The Linghou deposit in South China is a representative stratiform Cu-polymetallic deposit. The hydrothermal apatite associated with copper mineralization yields a lower intercept age of 161.0 ± 4.6 Ma, consistent with U–Pb ages from granodiorite porphyry intrusions, confirms that the Linghou deposit formed during the Late Jurassic. Microtextural and thermometric analyses of chalcopyrite and sphalerite indicate formation temperatures of 255–303 °C, characteristic of magmatic-hydrothermal systems. The restricted $\delta^{34}\text{S}$ range of sulfides supports a genetic link to felsic magmatism. Based on the comprehensive geological characteristics of the deposit, these findings establish the Linghou deposit as a skarn Cu-polymetallic system. The study not only resolves the genetic controversy surrounding this deposit type, but also underscores the exploration potential for analogous magmatic-hydrothermal copper systems along the Qin-Hang metallogenic belt.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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